Chapter 12. Sample Surveys

Sampling designs and stratified estimation

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12.1 Basic concepts of sample surveys

Motivation. A well-designed sample beats a slow, costly census; it needs a precise population and awareness of errors.

12.1.1 What is a sample survey

To learn about a population we can survey all of it or only part of it and estimate the whole from that part.

Definition (Census and sample survey)

(a) Surveying the entire population is a **census** (complete enumeration).
(b) Drawing a sample from part of the population and inferring population characteristics from it is a **sample survey**.

A census is exact in theory but often impossible or more error-prone in practice: surveying every Korean teenager's attitudes would take years and tens of thousands of interviewers, and the results would be outdated. Opinion surveys use samples.

Definition (Populations, frames and units)

(a) The **target population** is the one we want to study, defined precisely (are students on leave, or graduate students, "university students"?).
(b) The **sampling frame** (**frame**) is the list or rule identifying the units available for sampling.
(c) The **sampled population** is the one actually covered by the frame; a telephone-directory frame for "all adults in Korea" excludes unlisted adults.
(d) A **sampling unit** is an element of the population that can be selected. In a survey of students' attitudes it is each student; in a survey of farm household incomes it is each household.
(e) The **sampling design** specifies which units are selected and how. It is chosen to suit the parameters to be estimated and the structure of the population, so that estimation is as efficient as possible.

12.1.2 Errors and bias

Definition (Sampling and nonsampling error)

(a) **Sampling error** is the statistical error that arises because we infer about the whole population from part of it.
(b) **Nonsampling error** is every other error: the gap between target and sampled populations, interviewer mistakes, poorly worded questionnaires, and so on.

Sampling error shrinks like $1/\sqrt{n}$; nonsampling error does not depend on n (and can be larger in a census), so total error stops decreasing beyond some n.

Sampling error is computable and controlled through n; nonsampling error must be reduced by care in planning, collection and analysis. Its main sources:

(i) **Faulty questions**: unclear or leading questions distort answers.

(ii) **Wrong answers and interviewer inexperience**: incorrect or misunderstood answers (**response error**), or careless interviewers.

(iii) **Nonresponse error** (nonresponse bias): selected people do not respond. A low response rate makes the sample unrepresentative, especially when nonrespondents have particular tendencies.

(iv) **Coverage error**: some units are missing from (or wrongly included in) the frame, or the frame differs greatly from the target population.

(v) **Processing error**: errors in recording or in computer entry.

Besides $E(\hat{\theta})-\theta$, survey **bias** covers systematic distortions from nonsampling errors; its most serious source is a flawed design.

Case study: the 2015 UK general election polls

Before the 2015 UK election, five weeks of polls predicted no Commons majority, with Labour and the Conservatives level (the Conservatives had not won a majority since 1992). In fact the Conservatives won 331 seats, 91 more than Labour. The British Polling Council's inquiry found two main causes.
(i) *Wrongly chosen sampled population.* Frames over-represented Labour supporters (urban workers who answer surveys easily) and missed older Conservative voters.
(ii) *Neglect of nonresponse.* Respondents differed from nonrespondents. Conservative supporters likely had a lower response rate, so their real support was underestimated.

Similar errors: national attitudes surveyed through parents at particular schools, or landline-only telephone polls.

12.2 Sampling methods

Motivation. Simple random, systematic, stratified, cluster and stratified cluster sampling trade off cost and precision.

12.2.1 Simple random sampling

Definition (Simple random sampling)

Drawing n of N units so that all $\binom{N}{n}$ samples are equally likely is **simple random sampling** (SRS). In practice:
(i) number the units 1 to N;
(ii) select n of the numbers at random.

A random number generator is normally used. For a small population we can also draw lots from shuffled slips of paper.

Theorem (Sample mean under SRS without replacement)

Let a finite population have values $x_1,\dots,x_N$ with mean $\mu=\frac{1}{N}\sum_k x_k$ and variance $\sigma^2=\frac{1}{N}\sum_k(x_k-\mu)^2$. For an SRS of size n without replacement with sample mean $\overline{X}$ and sample variance S^2,

$$E(\overline{X})=\mu,\qquad \operatorname{Var}(\overline{X})=\frac{N-n}{N-1}\cdot\frac{\sigma^2}{n},\qquad E(S^2)=\frac{N}{N-1}\sigma^2.$$

Consequently $\left(1-\dfrac{n}{N}\right)\dfrac{S^2}{n}$ is an unbiased estimator of $\operatorname{Var}(\overline{X})$.

Proof. [Supplement]

Step 1. Let $\delta_k = 1$ if unit k is in the sample and $\delta_k = 0$ otherwise, so that $\overline{X} = \frac{1}{n}\sum_{k=1}^{N} x_k\delta_k$. By symmetry of SRS, for $k \neq l$,

$$P(\delta_k = 1) = \frac{n}{N}, \qquad P(\delta_k = \delta_l = 1) = \frac{n(n-1)}{N(N-1)}.$$

Hence $E(\overline{X}) = \frac{1}{n}\sum_k x_k \frac{n}{N} = \mu$.

Step 2. From Step 1,

$$\mathrm{Var}(\delta_k) = \frac{n}{N}\Big(1 - \frac{n}{N}\Big) = \frac{n(N-n)}{N^2}, \qquad \mathrm{Cov}(\delta_k, \delta_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3. Since $\sum_k \delta_k = n$ is constant, $\sum_k x_k\delta_k$ and $\sum_k d_k\delta_k$ (with $d_k = x_k - \mu$) have the same variance. Using $\sum_k d_k = 0$, so that $\sum_{k\neq l} d_k d_l = -\sum_k d_k^2 = -N\sigma^2$,

$$\begin{aligned}\mathrm{Var}\Big(\sum_k d_k\delta_k\Big) &= \frac{n(N-n)}{N^2}\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big] = \frac{n(N-n)}{N^2}\, N\sigma^2\Big(1 + \frac{1}{N-1}\Big) \\ &= \frac{n(N-n)}{N-1}\,\sigma^2.\end{aligned}$$

Dividing by n^2 gives $\mathrm{Var}(\overline{X}) = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n}$.

Step 4. Let $X_1, \ldots, X_n$ be the sample values. Each X_i has mean μ and variance σ^2, and

$$E\sum_{i=1}^{n}(X_i - \overline{X})^2 = E\sum_{i=1}^{n}(X_i - \mu)^2 - n\,E(\overline{X} - \mu)^2 = n\sigma^2 - \frac{N-n}{N-1}\sigma^2 = \frac{N(n-1)}{N-1}\sigma^2.$$

So $E(S^2) = \frac{N}{N-1}\sigma^2$. Finally,

$$E\Big[\Big(1 - \frac{n}{N}\Big)\frac{S^2}{n}\Big] = \frac{N-n}{N}\cdot\frac{N}{N-1}\cdot\frac{\sigma^2}{n} = \mathrm{Var}(\overline{X}).$$

■

12.2.2 Systematic sampling

Definition (Systematic sampling)

Systematic sampling takes a random unit among the first $k = N/n$ (the **sampling interval**) and every k-th unit thereafter.

E.g. $n = 200$ from $N = 10000$: $k = 50$, and start 23 gives units $23, 73, 123, \ldots, 9973$. It is simpler than SRS and spreads the sample evenly.

Exit polls use systematic sampling with k fixed in advance, since N and n are unknown; the joint exit polls of the three Korean broadcasters (2022 presidential, 2024 National Assembly elections) used $k = 5$.

Systematic sampling is convenient and is common in large surveys. If the list has a periodic pattern whose period matches k, however, it can produce serious bias.

12.2.3 Stratified random sampling

Definition (Stratified random sampling)

Drawing an SRS within each of several groups (**strata**) and pooling them is **stratified (random) sampling**.

Strata homogeneous within and heterogeneous across ensure every kind of unit appears, raising precision; stratification is the theoretically most important design.

E.g. if students' attitudes differ by college and year but are alike within them, small samples per stratum describe each accurately and precision improves greatly.

12.2.4 Cluster sampling

Definition (Cluster sampling)

Cluster sampling chooses some **clusters** (subpopulations) by SRS and surveys all their units (**single-stage**) or an SRS within each (**two-stage cluster sampling**).

It works well when clusters resemble each other and the population, and is cheaper: if attitudes depend on year but not major, survey all or some students of a few random departments.

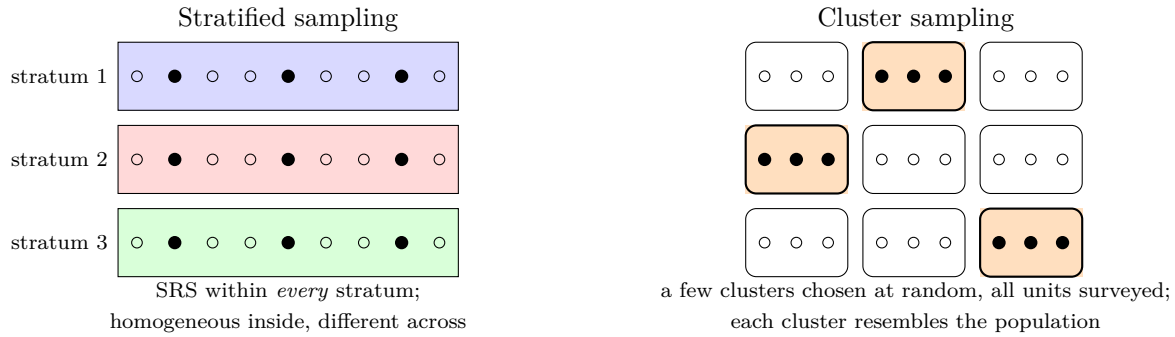


Figure 12.1: Stratified sampling versus (single-stage) cluster sampling; filled dots are sampled units.

12.2.5 Stratified cluster sampling

Definition (Stratified cluster sampling)

Stratified cluster sampling draws clusters (primary units) within each stratum and surveys them completely (**stratified single-stage cluster sampling**) or by SRS of secondary units (**stratified two-stage cluster sampling**).

E.g. with attitudes differing between colleges but little between departments: colleges as strata, departments as clusters, students drawn by SRS from chosen departments.

12.3 Estimation under stratified random sampling

Motivation. Estimators, standard errors and intervals for means and proportions under stratified sampling.

A population of N units has h strata of sizes N_i; **proportional allocation** splits n in proportion to stratum size:

$$n_i = n\Big(\frac{N_i}{N}\Big) \qquad (i = 1, 2, \ldots, h).$$

For means, let μ_i and σ_i^2 be the mean and variance of stratum i, with $\sigma_i^2 = \frac{1}{N_i}\sum(x-\mu_i)^2$ over the stratum. For proportions, let P_i be the proportion of units in stratum i with the attribute.

Definition 12-1 (Estimation of the population mean)

(i) Estimator:

$$\overline{X}_{\text{st}} = \frac{N_1}{N}\overline{X}_1 + \frac{N_2}{N}\overline{X}_2 + \cdots + \frac{N_h}{N}\overline{X}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \overline{X}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{\sigma_i^2}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\operatorname{SE}}(\overline{X}_{\text{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(1 - \frac{n_i}{N_i}\Big)\frac{S_i^2}{n_i}}\,.$$

Here $\overline{X}_i$ and S_i^2 are the sample mean and sample variance of stratum i, and the subscript st stands for "stratified".

Justification. [Supplement] The samples from different strata are independent SRSs without replacement.

Step 1. By the theorem of Section 12.2.1, $E(\overline{X}_i) = \mu_i$. Hence $E(\overline{X}_{\text{st}}) = \frac{1}{N}\sum_i N_i\mu_i = \mu$, so $\overline{X}_{\text{st}}$ is unbiased.

Step 2. By independence across strata and the same theorem,

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_i N_i^2 \operatorname{Var}(\overline{X}_i) = \frac{1}{N^2}\sum_i N_i^2 \frac{N_i - n_i}{N_i - 1}\cdot\frac{\sigma_i^2}{n_i},$$

which is (ii).

Step 3. The same theorem shows that $(1 - n_i/N_i)S_i^2/n_i$ is unbiased for $\operatorname{Var}(\overline{X}_i)$. Summing these estimates with weights N_i^2/N^2 gives an unbiased estimator of $\operatorname{Var}(\overline{X}_{\text{st}})$, and its square root is (iii). ■

Definition 12-2 (Estimation of the population proportion)

(i) Estimator:

$$\hat{P}_{\text{st}} = \frac{N_1}{N}\hat{P}_1 + \frac{N_2}{N}\hat{P}_2 + \cdots + \frac{N_h}{N}\hat{P}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \hat{P}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\hat{P}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{P_i(1-P_i)}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\operatorname{SE}}(\hat{P}_{\text{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(1 - \frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i}}\,.$$

Here $\hat{P}_i = X_i/n_i$ is the sample proportion of stratum i, where X_i is the number of sampled units in stratum i with the attribute.

Justification. [Supplement] Code each unit as 1 (has the attribute) or 0. Then $\mu_i = P_i$, $\sigma_i^2 = P_i(1-P_i)$ and $\overline{X}_i = \hat{P}_i$, so (i) and (ii) are the special cases of Definition 12-1.

For the sample variance of 0–1 data, $S_i^2 = \frac{n_i}{n_i-1}\hat{P}_i(1-\hat{P}_i)$. The exactly unbiased variance estimate is therefore

$$\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i-1}.$$

Formula (iii) with n_i in the denominator differs from it by the factor $(n_i-1)/n_i \approx 1$. Example 12-2 uses the exact version with $n_i - 1$. ■

Finite population correction and confidence intervals

$\frac{N_i-n_i}{N_i-1}$ in part (ii) of Definitions 12-1, 12-2 is the **finite population correction factor**; the estimated standard error uses $\frac{N_i-n_i}{N_i} = 1-\frac{n_i}{N_i}$ for unbiasedness (Section 12.2.1), and part (iii) replaces σ_i^2, P_i by S_i^2, $\hat{P}_i$.

The 95% confidence intervals for the mean and the proportion are

$$\bar{x}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = \bar{x}_{\mathrm{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{s_i^2}{n_i}},$$

$$\hat{p}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}}) = \hat{p}_{\mathrm{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{p}_i(1-\hat{p}_i)}{n_i}}.$$

The half-widths $z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}})$ and $z_{0.025}\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}})$ are called the **margin of error**, also known as the error limit or allowable error.

Example 12-1 (Household living expenses)

To survey monthly living expenses per household in a small city, the city was divided by standard of living into 3 regions (strata), and a stratified sample gave the following data (unit: 100,000 won).

Stratum	N_i	n_i	$\bar{x}_i$	s_i^2
1	2,800	30	18.5	12.2
2	5,600	60	15.2	11.3
3	3,400	30	10.7	10.5
Total	$N = 11{,}800$	$n = 120$		

Estimate the mean monthly living expenses per household and give a 95% confidence interval.

Solution. The estimate is

$$\bar{x}_{\mathrm{st}} = \frac{1}{11800}(2800\times 18.5 + 5600\times 15.2 + 3400\times 10.7) = 14.69.$$

The estimated variance is

$$\widehat{\mathrm{Var}}(\overline{X}_{\mathrm{st}}) = \frac{1}{11800^2}\Big\{2800^2\Big(1-\frac{30}{2800}\Big)\Big(\frac{12.2}{30}\Big) + 5600^2\Big(1-\frac{60}{5600}\Big)\Big(\frac{11.3}{60}\Big) + 3400^2\Big(1-\frac{30}{3400}\Big)\Big(\frac{10.5}{30}\Big)\Big\} = 0.09342.$$

So $\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = \sqrt{0.09342} = 0.3056$, and

$$\bar{x}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = 14.69 \pm 1.96\times 0.3056 = 14.69 \pm 0.60.$$

The 95% confidence interval is (14.09, 15.29), i.e. about 1.41 to 1.53 million won.

Example 12-2 (Support for a university policy)

A university grouped similar colleges into 3 strata A, B, C and surveyed students on leave about a university policy by stratified sampling.

College (stratum)	N_i	n_i	$\hat{p}_i$
A	3,500	35	0.56
B	4,500	45	0.60
C	2,000	20	0.40

Estimate the support rate and give a 95% confidence interval.
Solution. The estimate is

$$\hat{p}_{\text{st}} = \frac{1}{10000}(3500 \times 0.56 + 4500 \times 0.60 + 2000 \times 0.40) = 0.546.$$

Using $n_i - 1$ in the denominators,

$$\widehat{\text{Var}}(\hat{P}_{\text{st}}) = \frac{1}{10000^2}\Big\{3500^2\Big(1 - \frac{35}{3500}\Big)\Big(\frac{0.56 \times 0.44}{35-1}\Big) + 4500^2\Big(1 - \frac{45}{4500}\Big)\Big(\frac{0.60 \times 0.40}{45-1}\Big) + 2000^2\Big(1 - \frac{20}{2000}\Big)\Big(\frac{0.40 \times 0.60}{20-1}\Big)\Big\} = 0.00247.$$

So $\widehat{\text{SE}}(\hat{P}_{\text{st}}) = \sqrt{0.00247} = 0.0497$, and

$$\hat{p}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\hat{P}_{\text{st}}) = 0.546 \pm 1.96 \times 0.0497 = 0.546 \pm 0.097.$$

The 95% confidence interval is (0.449, 0.643), i.e. 44.9% to 64.3%. (The book centres the interval at 0.564 by a transposition slip and reports (0.467, 0.661); the margin of error 0.097 is the same.)

12.4 Problems in practical surveys

Motivation. A published margin of error covers only sampling error, not errors from misused randomness or poor questions.

A typical poll report: *automated wireless calls to virtual mobile numbers, sample 1,003, response rate* 2.3%, *sampling error* ±3.1 *points at* 95% *confidence.* The 3.1% is the margin of error of Section 12.3, covering sampling error only. Professional firms keep nonsampling error small, so it is then a good guide; in non-expert surveys nonsampling error can grow sharply, typically through misunderstood randomness and poor wording.

12.4.1 The concept of randomness

Random sampling selects units with probabilities fixed in advance; samples not drawn from the whole population by statistical theory incur severe bias (Section 12.1.2).

E.g. a student studying "generational differences in attitudes to household chores and gender equality" hands questionnaires to high-school students in one apartment complex for their parents. Childless families and the elderly are excluded, and one complex's households share socio-economic traits, so the sampled population differs greatly from the target.

The 2015 UK polls (Section 12.1) failed the same way: "random" read as "haphazard", or convenience chosen over design.

12.4.2 Writing the questionnaire

Questions must neither lead respondents to a particular answer nor confuse them with vague wording.

(a) *Leading questions.* "Should ancestral rites be kept in the traditional way, to honour the ancestors and teach tradition to our children?" and "Should they be simplified to fit changing values?" both invite "yes"; asked separately, the two proportions may sum to over 100%.

(b) *Ambiguous questions.* "Are you satisfied with your current life?" leaves "current life" undefined; ask about several aspects (work, family, leisure, friends) and combine. In "Do you have many friends?" "many" may mean 3 to one person and not 10 to another.

(c) *Campaigning disguised as a poll.* "Do you know that candidate X devoted his life to the democracy movement?" promotes a candidate and abuses surveys.

(d) *Pretesting.* A **pilot survey** finds such problems: asking students "monthly leisure budget? (1) under 100,000 won (2) 100–500 thousand (3) 500 thousand–1 million (4) over 1 million" yields mostly (1), revealing nothing about the distribution; revise the categories.

(e) *Length.* With too many questions, respondents tire and answer carelessly. Limit the number of items, and offering fair compensation for the respondent's time also helps.

Every item must be reviewed carefully. However sophisticated the sampling design, a badly written questionnaire yields distorted information.